

5.34 (*Recursive Exponentiation*) Write a recursive function `power(base, exponent)` that when invoked returns

$$\text{base}^{\text{exponent}}$$

For example, `power(3, 4) = 3 * 3 * 3 * 3`. Assume that `exponent` is an integer greater than or equal to 1. *Hint:* The recursion step would use the relationship

$$\text{base}^{\text{exponent}} = \text{base} * \text{base}^{\text{exponent}-1}$$

and the terminating condition occurs when `exponent` is equal to 1 because

$$\text{base}^1 = \text{base}$$